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SUPERSEDING
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May 19, 1958

FEDERAL SPECIFICATION

LAMP, INCANDESCENT, (SEALED-REFLECTOR, GENERAL SPECIFICATION FOR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—Lamps covered by this specification contain incandescent filament sources within a sealed inclosure. One wall of the inclosure is coated with a reflective surface and is shaped so as to direct the light flux through the opposite wall which may be made of clear glass or may be an optic to diffuse or otherwise redirect the light.

1.2 Classification.

1.2.1 *Types.*—Incandescent lamps covered by this specification shall be classified according to the following types:

Type I. —General service.
Type II. —Aviation.
Type III.—Automotive.
Type IV.—Military vehicle.
Type V. —Signal.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated, Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Boxes, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-L-10547 — Liners, Case, Waterproof.

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Military Standards:

MIL–STD–129—Marking for Shipment.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.—The lamps shall be designed and made so that they will satisfactorily meet the lighting-service conditions. All lamps shall conform to the manufacturers' standard sizes and dimensions of bulbs and bases on file at the National Bureau of Standards, or as shown in the applicable detail specifications (see 3.8).

3.2 Life to burn-out.—The lamps furnished under this specification shall give an average total life to burn-out of not less than the total life specified in the applicable detail specification (see 3.8). Unless otherwise stated in the detail specification, life tests shall be conducted at design voltage.

3.3 Initial candlepower requirements.—The candlepower distribution shall be such as to comply with the requirements given on the applicable detail specifications (see 3.8).

3.4 Candlepower maintenance requirements. — At 70 percent of rated life, the lamps shall meet the requirements given in the applicable detail specifications (see 3.8).

3.5 Electrical requirement. — At design voltages, the lamps shall not exceed the power or current ratings given in the applicable detail specifications (see 3.8).

3.6 Identification marking. — Each lamp shall be marked with the following identifying information:

a. Manufacturer's name or trademark.
b. Trade number (or rated watts if no trade number has been assigned).
c. Rated volts or amperes, or circuit volts.

3.7 Workmanship and materials. — The manufacturer is given the widest range in the selection of materials and processes of manufacture in order that sealed reflector lamps of maximum quality may be produced.

3.8 Detail requirements. — The detail requirements for the particular type lamps shall be as specified in the detail specifications.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Test procedure. — Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 All photometric and electrical measurements shall be measured at the design voltage of the lamps as given in the applicable detail specification.

4.2 Submission for inspection. — Lamps shall not be submitted for inspection prior to 5 days from the time that they are manufactured.

4.3 Lot definition.—A lot shall be all of the lamps of any one trade designation offered for acceptance at any one time.

PHYSICAL INSPECTION TEST

4.4 Lamp selection and test quantity.—The number of lamps to be selected for physical inspection shall be not less than 5 percent (minimum 10 lamps) of the lot, taken at random and proportionately from the individual packages of the lot. Lamps thus selected for inspection are known as the "inspection-test lamps".

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4.5 Rejection for minor physical defects.—Minor physical defects are those which are incompatible with good workmanship or good service. Minor defects include such things as bent contacts, slight discolorization of the lens or reflective coating, small bubbles in the glass, etc. A lot of lamps may be rejected without further test if 10 percent plus 1 lamp or more of the inspection-test lamps have minor physical defects.

4.6 Rejection for major physical defects.—Major physical defects are those which render lamps inoperative or which would unquestionably cause them to give poor service. Major defects include, but are not limited to, cracked glass, complete discolorization of the lens, peeled reflective coatings, and open or shorted filaments. A lot of lamps may be rejected without further test if 1 percent plus one lamp (2 percent plus 2 lamps when tested elsewhere than at the factory) have major physical defects. Individual lamps having major physical defects may be rejected and eliminated from an accepted lot, and shall be replaced by the manufacturer without additional charge.

4.7 Dimensions.—Any lamps which fail to conform to the dimensions specified in the applicable detail specification shall be considered to have major physical defects.

INITIAL RATING TEST

4.8 Lamp selection and test quantity.—Ten lamps shall be selected at random from each lot. These lamps may be from those taken for physical inspection tests (see 4.4) and are known as "rating-test lamps". The rating-test lamps shall be sufficiently burned or seasoned to reach stabilized values of wattage and candlepower before the initial rating test is conducted.

4.9 Rejection for deviation from candlepower requirements.—The entire lot may be rejected if the average of the ten rating-test lamps or more than two individual rating-test lamps fail to meet the candlepower requirements at any of the specified test points or at any point on the specified distribution curves.

4.10 Rejection for deviation from electrical requirements.—The entire lot of lamps may be rejected if the average power or current of the rating-test lamps exceeds the value given in the applicable detail specification by more than the tolerance shown.

LIFE PERFORMANCE TESTS

4.11 Lamp selection and test quantities.—After completion of the initial rating tests, the ten lamps shall be submitted to the National Bureau of Standards for life performance tests.

4.11.1 *Life-test lamps.*—The life-test lamps will be destroyed in the course of the life test. The inspector shall place a card in each container from which a life-test lamp has been selected, stating:

> "The lamp from this carton, required as part of the inspection of lamps in this package, has been removed and sent to the National Bureau of Standards, Washington, 25, D.C., for the required life test."

Lamps destroyed in life tests are to be counted as part of the quantity of lamps delivered by the contractor to the United States Government.

4.12 Lamp life.—The life of a lamp is the number of hours' life of the lamp to burn out. In the case of lamps that are life tested by intermittent operation, the lamp life shall be considered to be the total "ON" time.

4.13 Rejection for failure to meet life requirements.—A lot of lamps, provided that such lot is represented on test by 10 life-test lamps, may be rejected if the average life of the life-test lamps falls below the rated life as given in the applicable detail specifications by more than 25 percent.

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4.14 Determination of candlepower maintenance.—After the life-test lamps have been burned for 70 percent of their rated life, they shall be removed from the life-test racks and subjected to photometric tests to determine their compliance with the requirements of paragraph 3.4.

4.15 Rejection for failure to comply with candlepower maintenance requirements.—A lot of lamps, provided that such lot is represented by 10 life-test lamps, may be rejected if the lamps surviving at 70 percent of rated life fail to comply with the requirements of 3.4.

5. PREPARATION FOR DELIVERY

5.1 Requirements applicable to nonmilitary activities.

5.1.1 The levels of preservation, packaging and packing used are defined in Federal Standard No. 102.

5.1.2 *Packing.*—Unless otherwise specified, the subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.1.3 *Marking.* — Unless otherwise specified, the shipping containers shall be marked in accordance with Federal Standard No. 123.

5.2 Requirements applicable to Military activities.

5.2.1 *Preservation and packaging.* — The preservation and packaging shall be level A or C, as specified.

5.2.1.1 *Level A.*—Lamps furnished in the supplier's unit cartons shall be individually preserved in accordance with Military Specification MIL–P–116, Method 1C–2. Lamps of identical description shall be further intermediately packaged in fibreboard boxes conforming to Federal Specification PPP–B–636, type, class, and grade optional.

5.2.1.2 *Level C.*—Lamps shall be packaged in accordance with the supplier's standard practice.

5.2.2 *Packing.*—The packing shall be level A, B, or C as specified.

5.2.2.1 *Level A.*—Lamps shall be packed in close-fitting boxes conforming to Federal Specification PPP–B–636 (class 2), PPP–B–621 (class 2), PPP–B–601 (overseas type), or PPP–B–591 (overseas type). The boxes shall be lined with waterproof case liners conforming to Military Specification MIL–L–10547, sealed in accordance with the appendix thereto. Case liners shall not be provided for boxes conforming to Federal Specification PPP–B–636. The contents shall be cushioned, blocked and braced to prevent movement within the boxes. The gross weight shall not exceed the limitations of the applicable specification for fibreboard boxes and approximately 200 pounds for wood-cleated or wood boxes. Strapping shall be flat steel, type I, class B.

5.2.2.2 *Level B.*—Lamps shall be packed in close-fitting boxes conforming to Federal Specification PPP–B–636 (class 1), PPP–B–621 (class 1), PPP–B–601 (domestic type), or PPP–B–591 (domestic type). The contents shall be cushioned, blocked, and braced to prevent movement within the boxes.

5.2.2.3 *Level C.*—Packaged lamps of like description shall be packed in fiberboard boxes or containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.3 *Marking.*—In addition to any special marking required, packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

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6. NOTES

6.1 Ordering data. — Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Title, number, and date of detail specification covering lamp desired.

(*c*) Trade designation of lamp.

(*d*) Level of preservation and packaging, level of packing and marking required (See 5.1 and 5.2).

6.2 Transportation description. — Unless otherwise specified, the transportation description applicable to this commodity is:

Lamps, Electric, Incandescent, Including Photoflash.

Carload Minimum Weights:

15,000 pounds, subject to Rule 34, Uniform Freight Classification.

24,000 pounds, subject to Rule 34, Uniform Freight Classification. (See Notes 3 and 4, Items 55911 and 55912).

Motor Volume Minimum Weight, any quantity (see Note—Item 56290 of National Motor Freight Classification).

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Navy—A Sh S Y

Copies of this specification may be purchased for 5 cents each.

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